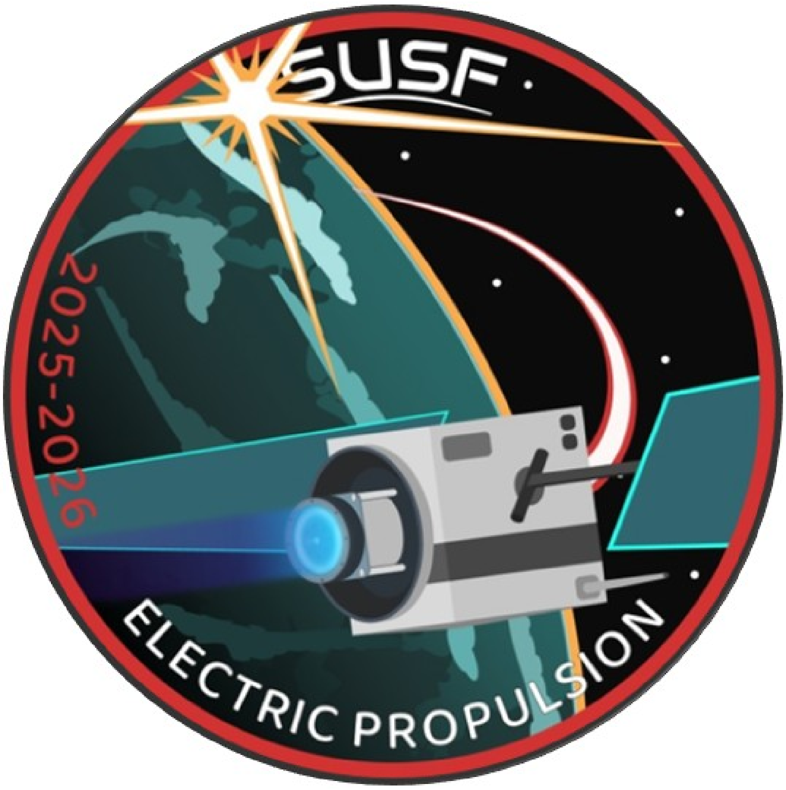
IO2 Documentation



Southampton University Electric Propulsion

March 2026

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Chapter 1

Introduction

Io2 is the natural successor of Io1 and builds directly from its legacy. Io1 was tested in February 2025 and was successfully ignited.

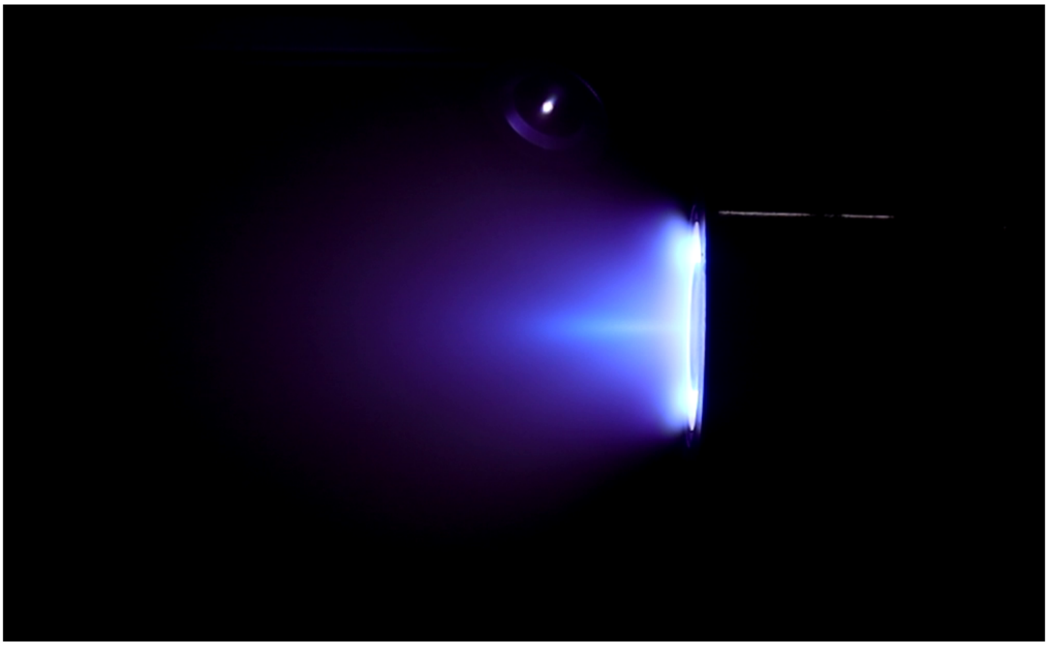


Figure 1.1: Io1 operating on Krypton gas at the David Fearne Electric Propulsion Laboratory.

The thruster was unable to operate steadily, and so we were unable to acquire hard data points. Despite this, we can still infer certain aspects of its performance. Based on prior magnetic field tests, the magnetic field is lopsided, resulting in a non ideal field topology, since for optimum performance, the field should be mostly radial. This directly resulted in the plume being quite divergent, as shown in Figure 1.1, likely causing many ions to have significant non-axial components in their velocities and not contributing to the overall thrust of the engine. It can also be seen that the plasma appears very dense close to the exit plane of the thruster, indicating that most of the ionisation and acceleration is occurring closer to the exit plane, which could explain instabilities and poor performance. Because of this, for Io2, great care was taken to ensure a highly radial field with magnetic lensing to ensure the greatest performance by keeping the magnetic field strength lower in the channel. In addition, an effort has been made to attempt magnetic shielding to reduce wall erosion, thermals and improve overall efficiency.

Table 1.1: Input design parameters for Io2.

Parameter	Value
Nominal Discharge Power	$1000W$
Nominal Discharge Voltage	$300V$
Propellant	Argon
Project Budget	£1600

Chapter 2

Channel Sizing

To size the cross-sectional discharge channel of a Hall effect thruster, it is common practice to utilise scaling laws. Often, a database containing geometric parameters along with performance metrics is used; in our case, we used Lee's database (1). The scaling laws derived can be used to define the mean channel diameter and channel width.

$$P = C_P d^2 V_d \tag{2.1}$$

$$h = C_{hd} d \tag{2.2}$$

In Equations 2.1, 2.2: P is the discharge power, C_P is the power scaling coefficient, d is the mean channel diameter $d = (d_0 + d_i)/2$, where d_i is the inside diameter, d_o is the outside diameter, V_d is the discharge voltage, h is the channel width and C_{hd} is the channel scaling coefficient.

The database used by Lee assumes all thrusters are operating on xenon and that in the centre of the channels have a peak magnetic field strength of $200G$; From this database, scaling coefficients were defined as $C_P = 633$ and $C_{hd} = 0.242$. Using the scaling coefficients, the channel variables are calculated to be $d = 72.6mm$ and $h = 17.6mm$ when using Table 1.1's parameters.

Thomas Munro shows the channel length to be inversely proportional to the first ionisation reaction rate coefficient in (2), in which he uses an analytical method to determine this quantity and relates it to the channel length by the following expression.

$$\lambda_i = \frac{v_n}{n_n \langle \sigma_i \nu_e \rangle} \ll L \tag{2.3}$$

Where λ_i is the ionisation mean free path in mm, v_n is the neutral velocity, n_n is the neutral number density, $\langle \sigma_i \nu_e \rangle$ is the ionisation reaction rate, and L is the discharge channel length.

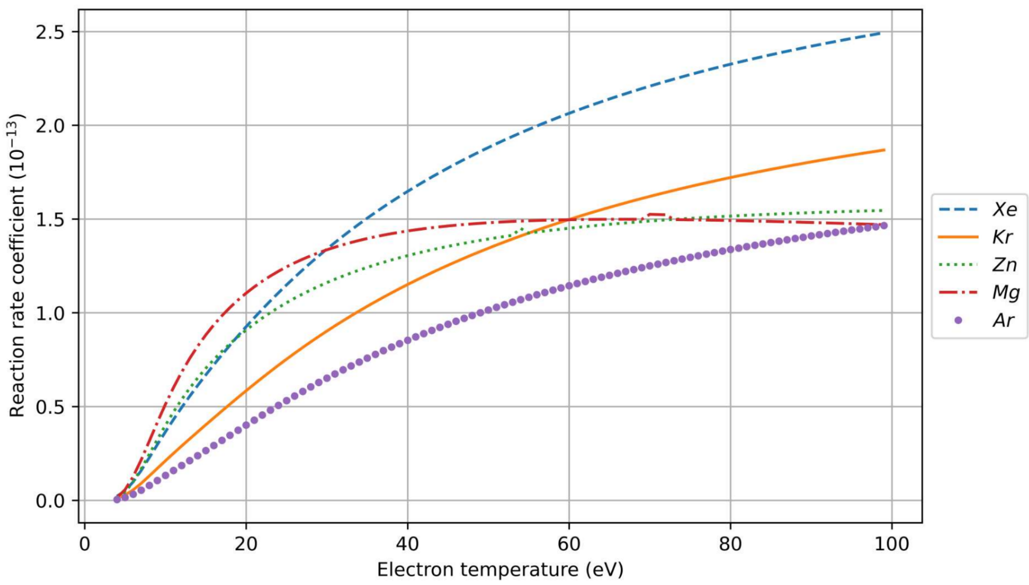


Figure 2.1: The calculated reaction rate coefficient against bulk electron temperature for propellants: Xenon, Krypton, Zinc, Magnesium and Argon (2)

$$T_{eV} = 0.12V_d \tag{2.4}$$

Using Figure 2.1 along with Equation 2.4, which shows the linear correlation between the averaged electron temperature, T_{eV}, in the discharge channel and the discharge voltage, V_d (3); An estimate of the reaction rate coefficient can be calculated. Taking the nominal discharge voltage along with several other assumptions made by Thomas in (2): $n_n \approx 1.2 \times 10^{19} m^{-3}$, which is an approximation for the number density of neutral particles in the discharge channel. The ionisation mean free path for argon can be calculated, $\lambda_i = 0.521mm$ where $T_{eV} = 36eV$. Comparing this to other thrusters from literature, it was decided to set the channel length, L to $35mm$, the channel is made longer than typical thrusters to give more time for ions to be formed, due to Argon's lower ionisation energy, while also satisfying Equation 2.3.

Chapter 3

Magnetic Field Design

The goal of the magnetic field topology is to "trap" electrons such that they can be used to ionise the propellant and then be accelerated, producing thrust. Traditional Hall effect thrusters have a topology such that the radial field component slowly rises from zero at the anode, rapidly increases in strength until peaking at the channel exit, or just before. The field strength then traditionally tapers off.

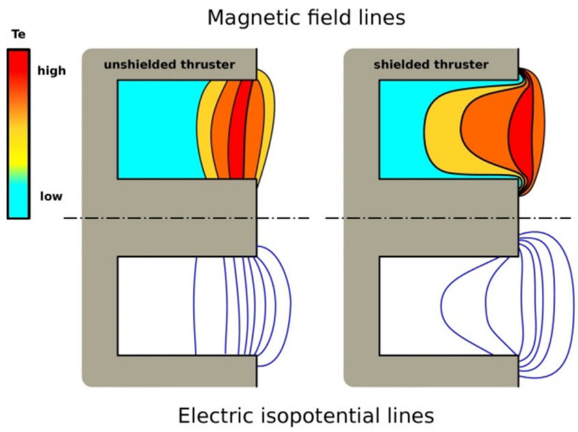


Figure 3.1: Illustration of the magnetic field isolines along with the electron temperature distribution in the channel for unshielded topologies (left) and shielded (right)().

Thrusters are typically magnetically shielded (MS) or unshielded (US). Magnetic shielding was developed as a method to extend the lifetimes of hall effect thrsuters by reducing the ion energy before it impacts the channel walls; This reduced erosion rates on the channel.

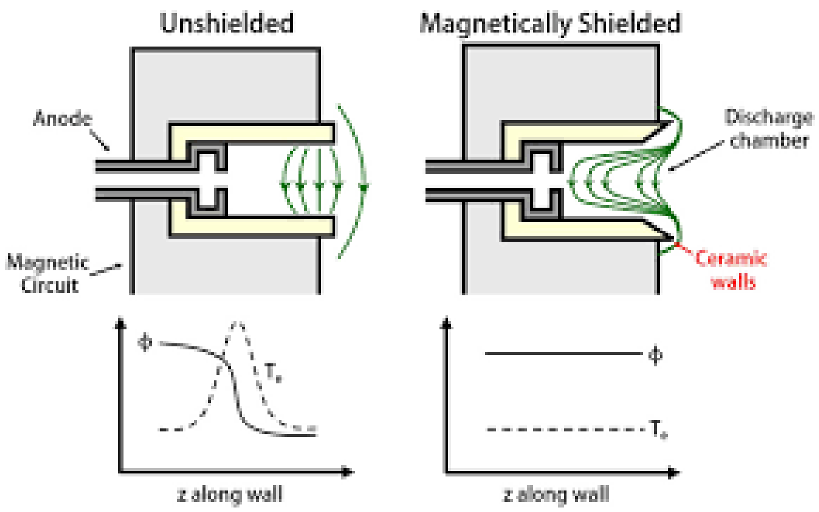


Figure 3.2: Illustration of the magnetic field isolines along with the electron temperature and potential distribution along the axial wall of the discharge channel for unshielded topologies (left) and shielded (right)().

Figure 3.1 and 3.2 show how the field lines curve into the channel for a MS thruster. It can also be seen that magnetic shielding significantly reduced the drop in potential towards the channel exit, this also displays that it can increase the propellant throughput capability of the thruster.

For Io2 the topology was designed with the intention of being magnetically shielded. A 2D cross section of the thruster was iterated using FEMM to simulate the magnetic field for different configurations of the magnetic circuit.

The final simulations to confirm the field topology were conducted using COMSOL Multiphysics's 3D magnetic simulations, with applied H-B curves for each material. This allowed for the adjustment of the channel geometry to account for inaccuracies in the FEMM simulations.

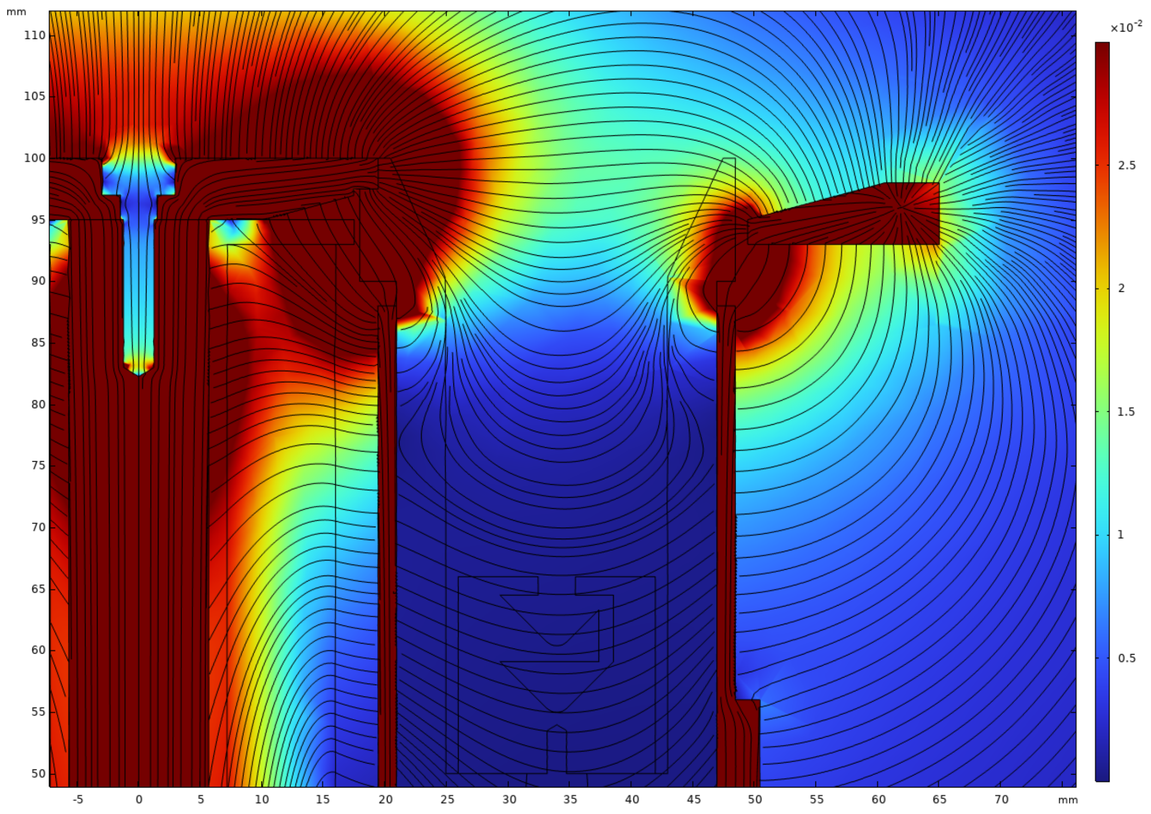


Figure 3.3: Magnetic Field Contour

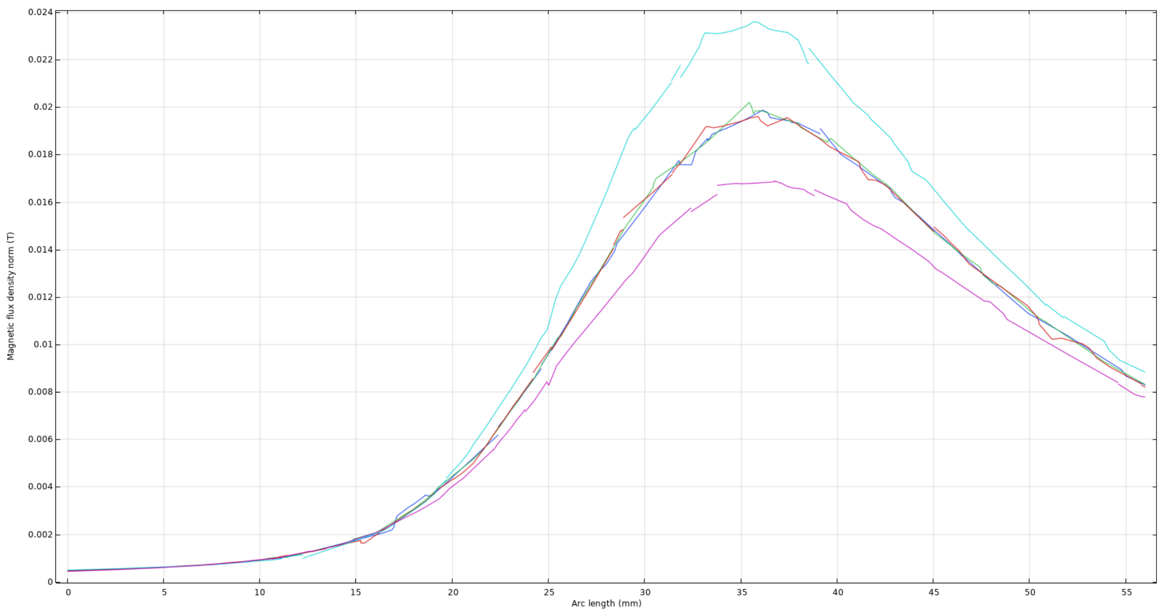


Figure 3.4: Magnetic Field Strength

The final field topology can be seen in Figure 3.3, and it is clear from here that the thruster was not MS. The reason for this was in the attempt to ensure that the peak field remained inside the discharge channel. To make the thruster MS, the magnetic screen walls could be raised higher, which would force the field to curve around the discharge channel. To bring the peak field inside the channel and be MS more advanced materials with higher saturating H-B curves would be needed, or a more creative design.

Figure 3.4 displays that the field strength peaked at the channel exit distance. The turquoise line is the field strength along the inside channel wall. The three plots that had an average peak field strength of $0.199T$ were taken at $5°$, $10°$ and $15°$ along the channel at the channel midpoint. The purple plot was taken along the outside channel wall.

Chapter 4

Anode & Gas Distributor Design

This component combines the function of the anode to produce the axial electric field along with that of the gas distributor to equally distribute the gas throughout the discharge channel. Since the flow rate of the gas is very low, it means that the number density of particles will also be low, which will result in few collisions over the discharge channel length. The collisionality of the gas can be described through the Knudsen number, K_n.

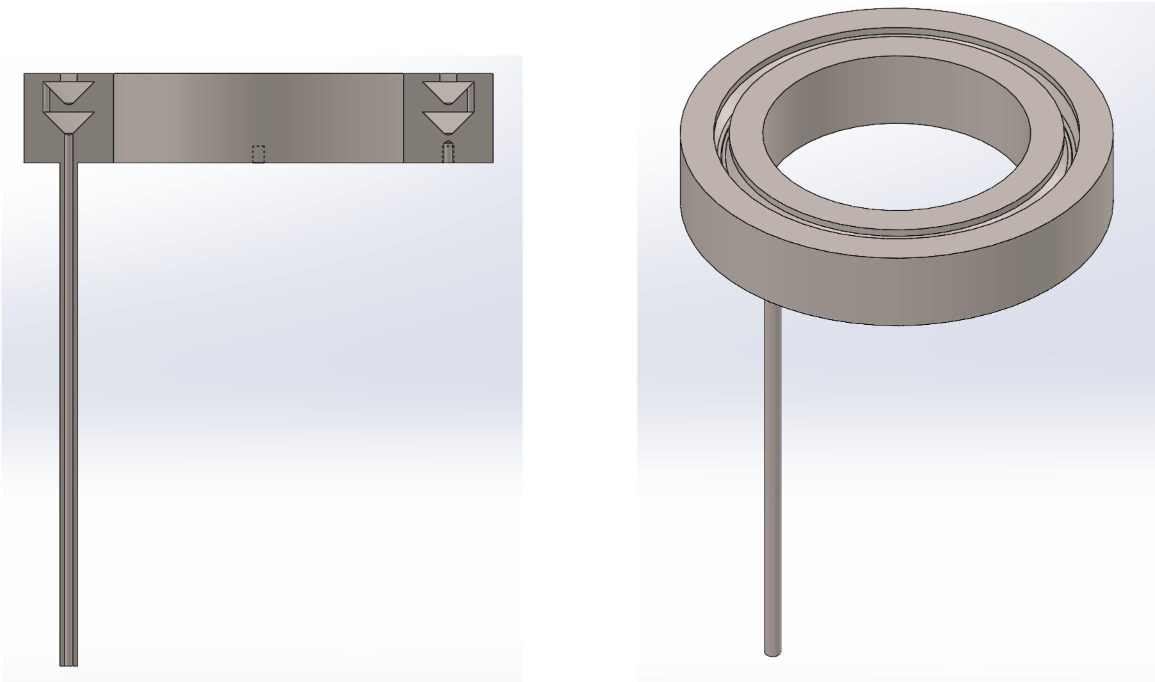
$$K_n = \frac{\lambda}{L} \tag{4.1}$$

In this equation, λ is the mean free path of a neutral atom travelling through the hall thruster, while L is a characteristic length. This could be the discharge channel length, the length of the feed pipe or the internal channels of the gas distributor. Depending on the value of the Knudsen number, different models are required to predict the flow of the gas. COMSOL gives an outline of when different models are appropriate in their Molecular flow user guide, which is summarised in Table 4.1

Table 4.1: Types of flow for different Knudsen number ranges (4).

Type of flow	Knudsen Number
Continuum flow	$K_n < 0.01$
Slip flow	$0.01 < K_n < 0.1$
Transitional flow	$0.1 < K_n < 10$
Free molecular flow	$K_n > 10$

To simulate the gas distribution throughout the channel, it was assumed that the $k - \omega$ model was appropriate, as in this compact, narrow component, the Knudsen number should be very small to maintain model accuracy. To move into the discharge channel, transitional flow models are likely needed, but as long as the flow uniformly exits the anode, there should be no reason to pursue further simulations downstream.

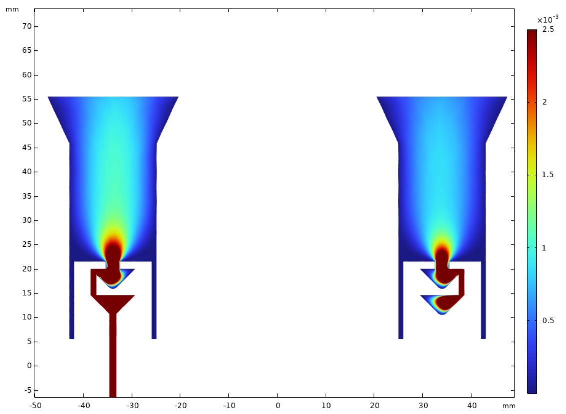


(a) (b)

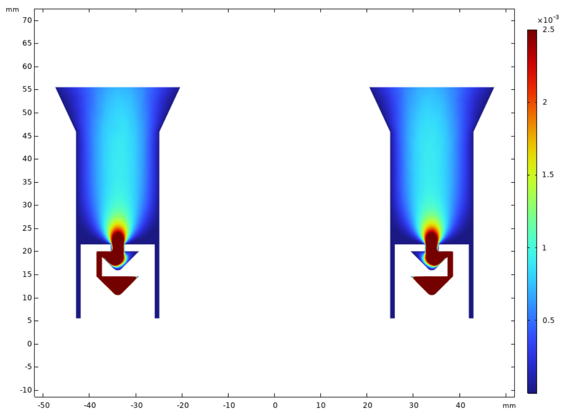
Figure 4.1: Anode geometry: (a) cross-section view, (b) isometric view.

Figure 4.1 shows the chosen anode, an 1/8 inch compression fitting will be fitted onto the inlet tube to then allow for 1/4 gas supply from the vacuum chamber. Gas will move through the tube and start being distributed by the first tubular ring before moving into the second, where it will hit baffles, further distributing it out before being emitted into the discharge channel.

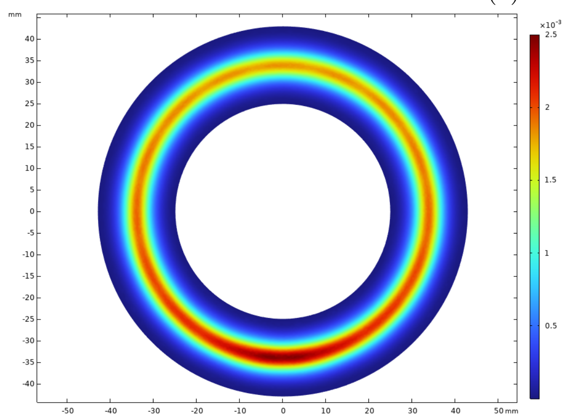
The anode also has 3 mounting holes, which take M2 screws to allow for easy attachment onto the thruster body so that it cannot be removed accidentally.



(a) XY plane view



(b) YZ plane view



(c) ZX plane view

Figure 4.2: Orthogonal COMSOL cross-sections of the anode geometry showing gas distribution pathways.

After iterating the design multiple times and checking the simulations as presented in Figure 4.2, the final design was settled upon. Figure 4.2 shows that the flow is uniformly discharged through the anode injector part of the gas distributor; the physics illustrated within the discharge channel is inaccurate and should not be considered to be accurate at all.

Chapter 5

Thermal Design

There are several power-loss mechanisms in a Hall-effect thruster; the most notable, which causes significant performance concerns, is thermal heating. To model the thermal loads on the hall thruster channel walls the following equation is given.

$$P_w = n_e e A_w \left[\left(\frac{kT_e}{e} \right)^{\frac{3}{2}} \left(\frac{2e}{\pi m_e} \right)^{\frac{1}{2}} \exp\left(\frac{e\varphi_S}{kT_e} \right) + \frac{1}{2} \left(\frac{kT_e}{m_i} \right)^{\frac{1}{2}} (\varepsilon - \varphi_S) \right] \tag{5.1}$$

Where n_e is the number density of electrons, e is the elementary charge, A_w is the surface area of the walls, k is the Boltzman constant, T_e is the electron temperature in Kelvin, m_e is the electron mass, φ_S is the sheath potential relative to the plasma, m_i is the ion mass, ε is the energy of the pre-sheath ion.

Discharge power to the anode can also be calculated and is another mechanism by which heat is transferred to the thruster.

$$P_a \approx 2T_{eV} I_d \tag{5.2}$$

Where T_{eV} is the temperature in electron-volts and I_d is the discharge current.

A model developed by Marconi C. Porto & et al. The model describes the total wall power losses to have constituents at the outer wall P_0, inner wall P_i, central wall P_c and edge wall P_e.

This relationships is describes through the following power balance in addition to the proceeding flux balance.

$$P_w = P_o + P_i + P_c + P_e \tag{5.3}$$

$$n\frac{P_o}{A_o} = \frac{P_i}{A_i} = \frac{P_c}{A_c} = \frac{P_e}{A_e} \quad n = 2 \tag{5.4}$$

Where A_x is a given area of a wall and x refers to subscripts o, i, c, e with identical meanings as before. As the magnetic field is stronger by the inner wall it is expected that a larger fraction of the plasma will be concentrated within this region because of this the authors (5) predict a power density of the order of $2:1$ is expected between the inner and outer walls.

Using Equation 5.1, 5.2, 5.3, 5.4 the discharge heating power is predicted in several regions summarised by Table 5.1.

Table 5.1: Table of areas and heating powers used for simulating temperature distribution around the thruster.

Region	Area $[mm^2]$	Power $[W]$
Total Wall	15791.16	211.58
Outer Wall	9912.08	96.78
Inner Wall	5445.52	106.34
Center edge	308.21	2.45
Outer edge	125.35	6.02

COMSOL 6.3 was used to thermally simulate the thruster using a solid conduction model with surface to surface radiation transfer. The required material parameters for each component are listed below in Table 5.2.

Table 5.2: Material table showing the thermal emissivity (ϵ), thermal conductivity (ω), density (ρ) and specific heat capacity (C_p) of each material used in the simulation

Material	ϵ	ω $[W/m^2]$	ρ $[kg/m^3]$	C_p $[J/(kg \cdot K)]$
1018 mild steel				
copper				
60xx aluminium				
HB boron nitride				

The resulting simulation gave the following temperature distribution.

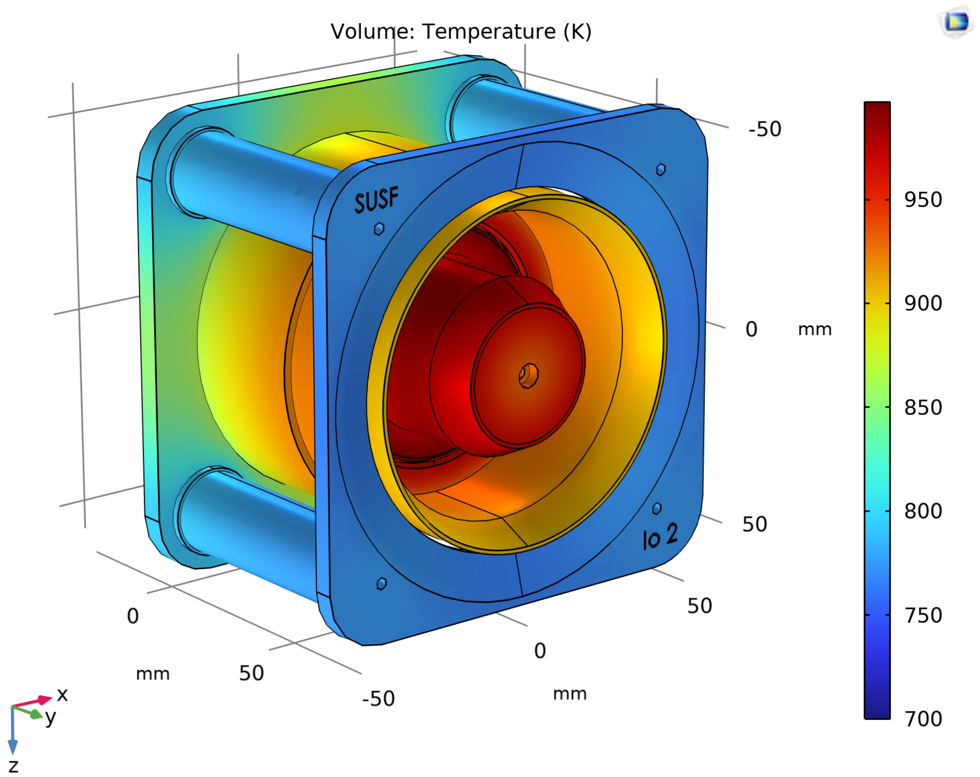


Figure 5.1: Temperature distribution for Io2.

The model does not account for the thruster stand or thrust balance so it is likely that temperatures will be much lower than are displayed in the simulation. All components remain within their operating temperature limits illustrating that the thruster should operate safely at the designed $1kW$, $300V$.

Bibliography

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[3] K. Dannenmayer and S. Mazouffre, "Elementary scaling relations for hall effect thrusters," *Journal of Propulsion and Power*, vol. 27, pp. 236–245, 01 2011.

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